FUNDAMENTALS OF ICT

COMPONENTS OF COMPUTER SYSTEM

1. HISTORY AND EVOLUTION OF COMPUTERS

The evolution of computers is generally divided into five distinct generations, each defined by the major technological breakthrough that changed how they operated.

First Generation (1940–1956): Vacuum Tubes The first generation utilized vacuum tubes for circuitry and magnetic drums for memory. These computers were characterized by their massive size, often taking up entire rooms, and were expensive to operate while generating significant amounts of heat. They relied on binary machine language and could only solve one problem at a time. Prominent examples include the ENIAC (Electronic Numerical Integrator and Computer) and the UNIVAC.

Second Generation (1956–1963): Transistors In this era, transistors replaced vacuum tubes, allowing computers to become smaller, faster, cheaper, and more energy-efficient. Operations moved from binary machine language to symbolic assembly language. High-level programming languages like COBOL and FORTRAN were also being developed during this time. Notable examples include the IBM 7094 and CDC 1604.

Third Generation (1964–1971): Integrated Circuits (ICs) Transistors were miniaturized and placed on silicon chips known as semiconductors or Integrated Circuits. This advancement drastically increased speed and efficiency. Instead of using punched cards and printouts, users began interacting via keyboards and monitors interfaced with an operating system, which allowed the device to run many different applications at once. Examples of this generation include the IBM 360 and PDP-11.

Fourth Generation (1971–Present): Microprocessors This generation is defined by the use of thousands of integrated circuits built onto a single silicon chip, known as Very Large Scale Integration (VLSI). Computers became small enough to fit on a desk, marking the birth of the Personal Computer (PC). The Intel 4004 chip, developed in 1971, located all the components of the computer (CPU, memory, input/output controls) on a minute chip. This era also saw the rise of the Internet, Graphical User Interfaces (GUIs), and the mouse. Examples include the Apple Macintosh, IBM PC, and modern smartphones.

Fifth Generation (Present and Beyond): Artificial Intelligence The current generation is based on Artificial Intelligence (AI), Ultra Large Scale Integration (ULSI), and Quantum Computing. These are devices that can respond to natural language input and are capable of learning and self-organization. The primary goal is to develop systems that simulate human intelligence. Examples include voice recognition systems like Siri and Alexa, as well as IBM Watson.

2. CLASSIFICATION OF COMPUTERS

Computers can be classified based on their size, processing power, and purpose.

Supercomputers - The fastest and most expensive computers available. • Used for specialized applications that require immense amounts of mathematical calculations, such as weather forecasting, nuclear energy research, and petroleum exploration. • Example: Fugaku, Summit.

Mainframe Computers - Very large and expensive computers capable of supporting hundreds or even thousands of users simultaneously. • Used by large organizations (banks, insurance companies) for bulk data processing, such as census data and transaction processing. • They focus on reliability and handling massive amounts of input/output data.

Minicomputers (Midrange Computers) - A class of multi-user computers that lies between the mainframe and the microcomputer. - Capable of supporting from 4 to about 200 simultaneous users. They were popular in the 1960s-80s but have largely been replaced by servers.

Microcomputers (Personal Computers) - Small, relatively inexpensive computers designed for an individual user. • They are based on the microprocessor technology. • Variations include: Desktops, Laptops, Tablets, and Smartphones.

3. COMPUTER HARDWARE AND SOFTWARE

A computer system consists of two primary elements: Hardware and Software.

A. Computer Hardware - refers to the physical, tangible parts of the computer system that you can touch.

• **Input Devices:** Tools used to send data to the computer. Examples: Keyboard, Mouse, Scanner, Microphone, Webcam.

• **Output Devices:** Tools that display or output the results of processed data. Examples: Monitor (Screen), Printer, Speakers, Projector.

• **System Unit / Processing:** The "brain" and housing of the computer.

- Motherboard: The main circuit board connecting all components.
- CPU (Central Processing Unit): Executes instructions.
- RAM (Random Access Memory): Temporary storage for active tasks.

• **Storage Devices:** Components used to store data permanently. Examples: Hard Disk Drive (HDD), Solid State Drive (SSD), USB Flash Drives, Memory Cards.

B. Computer Software Software refers to the set of instructions or programs that tell the hardware what to do. It cannot be touched physically.

• **System Software:** The software designed to run the computer's hardware and application programs. It acts as the interface between the hardware and the user.

- Operating Systems (OS): Windows, macOS, Linux, Android, iOS.
- Utility Programs: Antivirus, File Management tools, Disk Cleanup.

• **Application Software:** Programs designed to help the end-user perform specific tasks.

- General Purpose: Microsoft Word (word processing), Excel (spreadsheets), Web Browsers (Chrome, Edge).
- Specialized: Adobe Photoshop (graphic design), AutoCAD (engineering), Payroll Systems.